

Internship Task Submission

Full Name

Sabiha Abid

Applied Track/Domain

Web Development

Task Title

Interactive 3D Model Viewer

Project Description

This project is a web application that displays an interactive 3D model in the browser. Users can freely rotate and inspect the model from different angles through intuitive mouse interactions.

The application was developed with a focus on performance, responsiveness, and user experience while maintaining a clean and modern interface.

Live Demo: <https://sabiha-web-development-task.vercel.app/>

Approach and Work Completed

- Implemented rendering of a 3D model in the browser.
- Added interactive rotation controls for user interaction.
- Designed a responsive user interface that works across desktop and mobile devices.
- Optimized loading and rendering performance where possible.
- Tested the application on multiple screen sizes.

Assumptions

- Users access the application using a modern browser with WebGL support enabled.
- The selected 3D model is freely available for educational and demonstration purposes.

Limitations

- Only one 3D model is currently supported.
- Advanced interactions such as animations or model customization are not included.

Pending Work

- Additional camera controls.
- Support for multiple models.
- Performance optimizations for lower-end devices.

Tools and Resources Used

Development Tools

- OpenAI Codex / ChatGPT: AI coding assistant
- Visual Studio Code or local editor environment
- Node.js
- npm
- Vite
- Oxlint

Frontend Technologies

- HTML5
- CSS3
- JavaScript
- React
- WebGL

3D / Rendering Libraries

- Three.js
- React Three Fiber: [@react-three/fiber](#)
- Drei: [@react-three/drei](#)
- React Three Postprocessing: [@react-three/postprocessing](#)
- Postprocessing

Drei Features Used

- [OrbitControls](#)
- [Stars](#)
- [Environment](#)
- [Lightformer](#)
- [Loader](#)
- [useProgress](#)

Three.js Features Used

- Custom geometry
- BufferGeometry
- MeshPhysicalMaterial
- MeshDepthMaterial
- ShaderMaterial
- CanvasTexture
- Directional lights
- Ambient lights
- Shadows
- PCF soft shadow map
- Additive blending
- Procedural texture generation

Visual / Asset Resources

- Procedural Saturn surface texture generated in code
- Procedural Saturn ring texture generated in code
- Procedural sun glow generated in code
- Procedural nebula haze generated in code
- Local SVG icons and favicon in [/assets](#)
- No downloaded 3D models used
- No external image textures required for the main scene

Fonts

- Google Fonts:
 - Montserrat
 - Cormorant Garamond

References / Inspiration

- Pinterest
- NASA/Cassini Saturn imagery as visual inspiration
- Three.js documentation
- React Three Fiber documentation
- Drei documentation
- Vite documentation